

HE2003 Econometrics I

Tutorial 5 (Week 6) – Problems & Solutions

Academic Year 2026/2027, Semester 1

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Overview

- **Lecture coverage:** Lecture 5 — *unbiasedness and efficiency of $\bar{Y}$, the Gauss–Markov theorem for the OLS estimators, and the general condition for a linear estimator $\sum a_i Y_i$ to be unbiased*; plus Lecture 4’s decomposition $TSS = ESS + SSR$, used in Q2 to show $R^2 \leq 1$.
- **Questions:** 3 questions from Tutorial Question Set No. 5; solutions extend the official Tut 5 slides, whose Q1(c) derivation was delivered as an unrecovered equation image — reconstructed in full below.
- **Notation:** as in lectures — u_i population error, $\hat{u}_i$ OLS residual; ESS explained, SSR residual, $TSS = ESS + SSR$; “Least Squares Assumption 1/2/3” for the OLS assumptions (never “SLR.1–5”); critical value 1.96 for a two-sided 5% test.

Question 1 (Unbiasedness, efficiency, Gauss–Markov, and BLUE)

Problem

- Explain the notion of unbiasedness and efficiency by using the sample average $\bar{Y}$ as an example.
- Explain the Gauss–Markov Theorem for the OLS estimators.
- Find out the condition that guarantees the linear estimator $\sum_{i=1}^n a_i Y_i$ is an unbiased estimator of the population mean μ .
- Explain the example regarding the efficiency of $\bar{Y}$ on slides 6–8 of Lecture 5.

(Tutorial Question Set No. 5, Q1.)

Solution

Part (a): Unbiasedness and efficiency, via $\bar{Y}$

Definitions. Let $\hat{\theta}$ be an estimator of a population parameter θ , built from a random sample $Y_1, \dots, Y_n$ (so $\hat{\theta}$ is itself a random variable, with a *sampling distribution* across repeated samples). Then:

- $\hat{\theta}$ is **unbiased** if the mean (centre) of its sampling distribution equals the true parameter: $E[\hat{\theta}] = \theta$. On average, across repeated samples, the estimator gets the right answer.
- Given two unbiased candidates, $\hat{\theta}$ is **more efficient** than a competitor $\tilde{\theta}$ if it has the tighter sampling distribution, i.e. $\text{Var}[\hat{\theta}] < \text{Var}[\tilde{\theta}]$. Efficiency is only a meaningful comparison *among* unbiased estimators – a biased estimator can trivially have small variance while being centred on the wrong value.

Method 1: Official comparison – $\bar{Y}$ versus $\tilde{Y} = \frac{1}{2}(Y_1 + Y_n)$. Let $Y_1, \dots, Y_n$ be i.i.d. draws with $E[Y_i] = \mu$, $\text{Var}(Y_i) = \sigma^2$. Consider two candidate estimators of μ : the sample average $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$, and $\tilde{Y} = \frac{1}{2}(Y_1 + Y_n)$ (the average of the first and last observations only).

Unbiasedness. By linearity of expectation,

$$E[\bar{Y}] = E\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] = \frac{1}{n} \sum_{i=1}^n E[Y_i] = \frac{1}{n}(n\mu) = \mu,$$

$$E[\tilde{Y}] = E\left[\frac{1}{2}(Y_1 + Y_n)\right] = \frac{1}{2}(E[Y_1] + E[Y_n]) = \frac{1}{2}(\mu + \mu) = \mu.$$

Both are unbiased for μ .

Efficiency. Using independence across observations (so all cross covariances vanish),

$$\text{Var}[\bar{Y}] = \text{Var}\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Y_i) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n},$$

$$\text{Var}[\tilde{Y}] = \text{Var}\left[\frac{1}{2}(Y_1 + Y_n)\right] = \frac{1}{4}(\text{Var}(Y_1) + \text{Var}(Y_n)) = \frac{1}{4}(\sigma^2 + \sigma^2) = \frac{\sigma^2}{2},$$

where the second equality for $\tilde{Y}$ uses $\text{Cov}(Y_1, Y_n) = 0$ (independence). Comparing,

$$\boxed{\text{Var}(\bar{Y}) = \frac{\sigma^2}{n} < \text{Var}(\tilde{Y}) = \frac{\sigma^2}{2} \quad \text{for } n > 2,}$$

so $\bar{Y}$ is more efficient than $\tilde{Y}$ whenever the sample has more than two observations (for $n = 2$, $\tilde{Y} = \bar{Y}$ exactly, and the two coincide).

Beyond the slides: adding Y_1 alone as a third candidate. A single observation Y_1 is also an unbiased estimator: $E[Y_1] = \mu$ trivially, with $\text{Var}(Y_1) = \sigma^2$. Ranking all three estimators by variance for $n > 2$:

$$\boxed{\text{Var}(\bar{Y}) = \frac{\sigma^2}{n} < \text{Var}(\tilde{Y}) = \frac{\sigma^2}{2} < \text{Var}(Y_1) = \sigma^2.}$$

All three are unbiased, but $\bar{Y}$ – which uses *every* observation, not just one or two – is the most efficient. This is not a coincidence: part (c) below derives the exact condition under which a linear combination $\sum a_i Y_i$ is unbiased, and its extension shows that $\bar{Y}$ (i.e. $a_i = 1/n$ for all i) is in fact the *most* efficient linear unbiased estimator of μ among all such combinations – the BLUE property referenced in part (d) and generalised to the OLS slope/intercept in part (b).

Interpretation. Using more of the sample does not introduce bias – averaging preserves $E[\cdot] = \mu$ regardless of how many observations are included – but it does reduce variance, because averaging more independent draws cancels out more sampling noise. $\tilde{Y}$ wastes the information in $Y_2, \dots, Y_{n-1}$; Y_1 wastes even more. $\bar{Y}$ is the efficient choice precisely because it is the linear unbiased combination that uses all the sample information symmetrically.

Part (b): Gauss–Markov Theorem for the OLS estimators

Least Squares Assumptions. The three Least Squares Assumptions (LSA) for the simple regression $Y_i = \beta_0 + \beta_1 X_i + u_i$ are:

LSA1. $E[u_i | X_i] = 0$ (the error has conditional mean zero).

LSA2. (X_i, Y_i) , $i = 1, \dots, n$, are i.i.d. draws from their joint population distribution (simple random sampling).

LSA3. Large outliers are unlikely: X_i and Y_i have finite fourth moments.

LSA1–3 alone are enough to establish that $\hat{\beta}_0, \hat{\beta}_1$ are *consistent* and, in large samples, approximately normal – but they are **not** enough for Gauss–Markov’s stronger finite-sample efficiency claim.

The technical condition: homoskedasticity. The Gauss–Markov theorem additionally requires the *homoskedasticity-only* technical condition $\text{Var}(u_i | X_i) = \sigma_u^2$, a constant that does not depend on X_i . Under LSA1–3 *plus* homoskedasticity, the theorem states:

Gauss–Markov Theorem. Under the Least Squares Assumptions and homoskedasticity, the OLS estimators $\hat{\beta}_0, \hat{\beta}_1$ are the **Best Linear Unbiased Estimators (BLUE)** of β_0, β_1 . That is, for *any* alternative estimators $\tilde{\beta}_0, \tilde{\beta}_1$ that are (i) linear in $Y_1, \dots, Y_n$ and (ii) unbiased, $E[\tilde{\beta}_0] = \beta_0$, $E[\tilde{\beta}_1] = \beta_1$,

$$\text{Var}[\hat{\beta}_0] \leq \text{Var}[\tilde{\beta}_0], \quad \text{Var}[\hat{\beta}_1] \leq \text{Var}[\tilde{\beta}_1].$$

Unpacking BLUE.

- **B – Best:** smallest variance among the class being compared – i.e. most efficient.
- **L – Linear:** the estimator can be written as a linear function of $Y_1, \dots, Y_n$ (weights may depend on the X_i 's, but not on the Y_i 's). OLS itself is such a weighted sum, $\hat{\beta}_1 = \sum_i w_i Y_i$ with $w_i = (X_i - \bar{X}) / \sum_j (X_j - \bar{X})^2$.
- **U – Unbiased:** $E[\hat{\beta}_0] = \beta_0$, $E[\hat{\beta}_1] = \beta_1$.
- **E – Estimator:** a rule for computing an estimate from the sample.

What the theorem does and does not say.

- It *does* say: within the restricted class of estimators that are both linear in Y and unbiased, no competitor beats OLS on variance. Under LSA1–3 plus homoskedasticity, there is no point searching for another linear unbiased estimator – none will be more efficient.
- It does *not* say OLS is best among *all* estimators. A *biased* estimator (e.g. a shrinkage/ridge estimator) can have smaller mean squared error by trading a small amount of bias for a larger reduction in variance; Gauss–Markov says nothing about that trade-off.
- It does *not* require normality of u_i – the BLUE result is proved using only means, variances, and independence, exactly as in part (a).
- It does *not* hold as stated under heteroskedasticity: if $\text{Var}(u_i | X_i)$ varies with X_i , OLS remains unbiased (LSA1–3 are enough for that) but is no longer BLUE – a weighted least squares (GLS) estimator that down-weights high-variance observations can then achieve a smaller variance.

Part (c): unbiasedness condition for $\sum a_i Y_i$

Official slide gap: the Tut 5 slide shows only the connective phrases (“Assume $E[\cdot] = \mu \dots$ ”, “For this estimator to be unbiased...”, “Therefore, the required condition is...”) with the equations themselves missing (an unrecovered image). The full derivation is reconstructed below.

Method 1: Direct expectation

Step 1. Let $Y_1, \dots, Y_n$ have $E[Y_i] = \mu$ for every i , and let $a_1, \dots, a_n$ be constants (not random). Consider the linear estimator $\hat{\mu} = \sum_{i=1}^n a_i Y_i$. By linearity of expectation,

$$E[\hat{\mu}] = E\left[\sum_{i=1}^n a_i Y_i\right] = \sum_{i=1}^n a_i E[Y_i] = \sum_{i=1}^n a_i \mu = \mu \sum_{i=1}^n a_i.$$

Step 2. For $\hat{\mu}$ to be unbiased for μ we need $E[\hat{\mu}] = \mu$ to hold *for every possible value of μ* (the estimator must work whatever the true population mean is, not just for one particular μ). That is,

$$\mu \sum_{i=1}^n a_i = \mu \quad \text{for all } \mu \iff \left(\sum_{i=1}^n a_i - 1 \right) \mu = 0 \quad \text{for all } \mu \iff \sum_{i=1}^n a_i = 1.$$

(Taking, e.g., $\mu = 1$ already forces $\sum a_i = 1$; any other $\mu \neq 0$ confirms it.)

$$\hat{\mu} = \sum_{i=1}^n a_i Y_i \text{ is unbiased for } \mu \iff \sum_{i=1}^n a_i = 1.$$

Beyond the slides: variance, and why $a_i = 1/n$ is optimal. The slides stop at the unbiasedness condition; the natural next question – which choice of $(a_1, \dots, a_n)$, among all those satisfying $\sum a_i = 1$, minimises the variance – is not asked explicitly, but it is exactly what makes part (a)/(d)'s ranking rigorous. Under i.i.d. sampling with $\text{Var}(Y_i) = \sigma^2$ and independence,

$$\text{Var}(\hat{\mu}) = \text{Var}\left(\sum_{i=1}^n a_i Y_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(Y_i) = \sigma^2 \sum_{i=1}^n a_i^2.$$

We minimise $\sum a_i^2$ subject to the constraint $\sum a_i = 1$. Write $a_i = \frac{1}{n} + b_i$; the constraint becomes $\sum b_i = \sum a_i - 1 = 0$. Then

$$\sum_{i=1}^n a_i^2 = \sum_{i=1}^n \left(\frac{1}{n} + b_i \right)^2 = \sum_{i=1}^n \left[\frac{1}{n^2} + \frac{2b_i}{n} + b_i^2 \right] = \frac{1}{n} + \underbrace{\frac{2}{n} \sum_{i=1}^n b_i}_{=0} + \sum_{i=1}^n b_i^2 = \frac{1}{n} + \sum_{i=1}^n b_i^2,$$

i.e. the identity

$$\sum_{i=1}^n a_i^2 = \sum_{i=1}^n \left(a_i - \frac{1}{n} \right)^2 + \frac{1}{n} \geq \frac{1}{n},$$

with equality if and only if $a_i = 1/n$ for every i (since a sum of squares is zero only if every term is zero). Hence the minimiser of $\sum a_i^2$ subject to $\sum a_i = 1$ is $a_i = 1/n$ for all i , giving $\hat{\mu} = \bar{Y}$ and $\text{Var}(\hat{\mu})_{\min} = \sigma^2/n = \text{Var}(\bar{Y})$. This shows directly that

$$\bar{Y} \text{ is the Best Linear Unbiased Estimator (BLUE) of } \mu.$$

This is precisely the KEY CONCEPT stated (without proof) on Lecture 5 slide 5, $\text{Var}(\bar{Y}) \leq \text{Var}(\hat{Y})$ for any linear unbiased $\hat{Y} = \sum a_i Y_i$; the algebra above supplies the missing proof, and is the sample-mean analogue of the Gauss–Markov theorem proved (without full algebra) in part (b) for the OLS slope and intercept.

Interpretation. Only estimators whose weights sum to one can possibly be unbiased for μ – this rules out, for instance, $\sum a_i Y_i$ with all $a_i = 1/(n-1)$ (which would be biased, since it targets $n\mu/(n-1) \neq \mu$). Among the unbiased ones, spreading the weight equally across all n observations minimises variance; any unevenness in the weights (any $b_i \neq 0$) strictly increases $\sum a_i^2$ and hence the variance, without helping unbiasedness at all.

Part (d): the Lecture 5 slides 6–8 simulation

Setup. 20 independent random samples, each of size $n = 10$, are drawn from $N(\mu = 2, \sigma^2 = 1)$. From each sample two estimators of μ are computed: the first observation Y_1 , and the sample average $\bar{Y} = \frac{1}{10}(Y_1 + \dots + Y_{10})$.

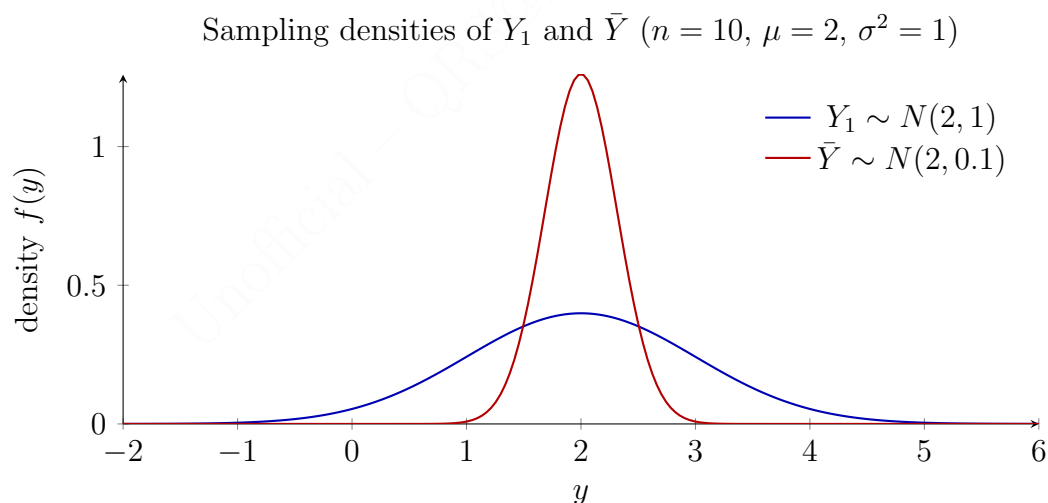
Simulation evidence.

- Across the 20 samples, Y_1 ranges over $[-0.64, 4.27]$, a spread of 4.91; $\bar{Y}$ ranges over the much tighter $[1.16, 2.58]$, a spread of only 1.42.
- In 16 of the 20 samples, $\bar{Y}$ lands closer to the true value $\mu = 2$ than Y_1 does.
- The average of Y_1 over the 20 simulations is about 1.89; the average of $\bar{Y}$ is about 1.96. Both are close to 2, which is the empirical signature of *unbiasedness*: repeated sampling and averaging the estimates recovers the true parameter for either estimator, even though any single sample's Y_1 can be far off.

Theoretical benchmark. With $\sigma^2 = 1$ and $n = 10$,

$$\text{Var}(Y_1) = \sigma^2 = 1, \quad \text{Var}(\bar{Y}) = \frac{\sigma^2}{n} = \frac{1}{10} = 0.1.$$

So $\text{Var}(Y_1)$ is exactly 10 times $\text{Var}(\bar{Y})$ – consistent with the visibly wider spread of Y_1 's simulated values above, and matching the general result $\text{Var}(\bar{Y}) = \sigma^2/n$ from part (a). The two sampling distributions are $Y_1 \sim N(2, 1)$ and $\bar{Y} \sim N(2, 0.1)$ (exact normality here because the population itself is normal):



Interpretation. The figure – redrawn here since the original Lecture 5 slides present only the numerical table, not this density picture – makes the efficiency comparison visual: both curves are centred at the true $\mu = 2$ (unbiasedness), but the $\bar{Y}$ curve is far more concentrated (efficiency). This is exactly the point of the simulation: unbiasedness alone does not pin down a “good” estimator, since Y_1 is unbiased too; efficiency is the criterion that separates $\bar{Y}$ from Y_1 , and by part (c) $\bar{Y}$ is in fact the most efficient linear unbiased estimator possible.

$$\text{Var}(Y_1) = 1, \quad \text{Var}(\bar{Y}) = 0.1; \quad \bar{Y} \text{ is unbiased and far more efficient than } Y_1.$$

Technique note. "Unbiased" is a statement about where the sampling distribution is centred, averaged over hypothetical repeated samples; it says nothing about how spread out a single draw from that distribution can be. Always report both properties together, and always prefer the estimator with smaller variance among unbiased candidates.

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Question 2 (Why R^2 cannot exceed 1)

Problem

In this question we show why R^2 cannot be larger than 1.

- (a) Show that $\text{TSS} = \sum_{i=1}^n [\hat{u}_i + (\hat{Y}_i - \bar{Y})]^2$.
- (b) Show that $\text{TSS} = \text{ESS} + \text{SSR}$, where TSS and ESS are as defined in the lecture slides and $\text{SSR} = \sum_{i=1}^n \hat{u}_i^2$. [Hint: use $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$ and $\sum_{i=1}^n \hat{u}_i = 0$.]
- (c) Show that $R^2 \leq 1$.

(Tutorial Question Set No. 5, Q2.)

Solution

Method 1: Algebraic decomposition (official route)

Part (a). By definition of the OLS residual, every observation decomposes as $Y_i = \hat{Y}_i + \hat{u}_i$, so

$$Y_i - \bar{Y} = (\hat{Y}_i + \hat{u}_i) - \bar{Y} = \hat{u}_i + (\hat{Y}_i - \bar{Y}).$$

Squaring and summing over $i = 1, \dots, n$ gives, directly,

$$\text{TSS} = \sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n [\hat{u}_i + (\hat{Y}_i - \bar{Y})]^2. \quad \blacksquare$$

Part (b). Expand the square from part (a):

$$\text{TSS} = \sum_{i=1}^n [\hat{u}_i + (\hat{Y}_i - \bar{Y})]^2 = \underbrace{\sum_{i=1}^n \hat{u}_i^2}_{=\text{SSR}} + \underbrace{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}_{=\text{ESS}} + 2 \sum_{i=1}^n \hat{u}_i (\hat{Y}_i - \bar{Y}).$$

It remains to show the cross term vanishes. Expanding it,

$$2 \sum_{i=1}^n \hat{u}_i (\hat{Y}_i - \bar{Y}) = 2 \left[\sum_{i=1}^n \hat{u}_i \hat{Y}_i - \bar{Y} \sum_{i=1}^n \hat{u}_i \right].$$

Both bracketed sums are the two classic residual properties from the OLS first-order conditions (Tutorial 3, Q2), re-derived here in one line each so the argument is self-contained. The OLS estimators $\hat{\beta}_0, \hat{\beta}_1$ solve $\min_{b_0, b_1} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2$; the two FOCs, evaluated at the optimum and divided by -2 , are

$$\sum_{i=1}^n \hat{u}_i = 0 \quad \text{and} \quad \sum_{i=1}^n \hat{u}_i X_i = 0,$$

where $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$. The first is exactly $\sum \hat{u}_i = 0$. For the second sum, substitute $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$:

$$\sum_{i=1}^n \hat{u}_i \hat{Y}_i = \sum_{i=1}^n \hat{u}_i (\hat{\beta}_0 + \hat{\beta}_1 X_i) = \hat{\beta}_0 \sum_{i=1}^n \hat{u}_i + \hat{\beta}_1 \sum_{i=1}^n \hat{u}_i X_i = \hat{\beta}_0(0) + \hat{\beta}_1(0) = 0.$$

Hence the cross term is $2[0 - \bar{Y} \cdot 0] = 0$, and

$$\boxed{\text{TSS} = \text{SSR} + \text{ESS} = \text{ESS} + \text{SSR}. \quad \blacksquare}$$

(Rearranged, this also gives $R^2 = \text{ESS}/\text{TSS} = (\text{TSS} - \text{SSR})/\text{TSS} = 1 - \text{SSR}/\text{TSS}$, used in part (c).)

Part (c). $\text{SSR} = \sum_{i=1}^n \hat{u}_i^2$ is a sum of squares, so $\text{SSR} \geq 0$. From part (b), $\text{TSS} = \text{ESS} + \text{SSR}$, so

$$\text{ESS} = \text{TSS} - \text{SSR} \leq \text{TSS} \quad (\text{since } \text{SSR} \geq 0).$$

Provided $\text{TSS} > 0$ (i.e. Y is not degenerate, all equal to $\bar{Y}$), divide through by TSS :

$$R^2 = \frac{\text{ESS}}{\text{TSS}} \leq 1.$$

Equivalently, using $R^2 = 1 - \text{SSR}/\text{TSS}$ from part (b) and $\text{SSR} \geq 0$, $\text{TSS} \geq 0$: since $0 \leq \text{SSR} \leq \text{TSS}$, it follows that $0 \leq \text{SSR}/\text{TSS} \leq 1$, hence

$$\boxed{0 \leq R^2 = 1 - \frac{\text{SSR}}{\text{TSS}} \leq 1.}$$

($\text{ESS} = \sum (\hat{Y}_i - \bar{Y})^2 \geq 0$ similarly gives the lower bound $R^2 \geq 0$ directly.) $\blacksquare$

Method 2: Geometric view (Pythagoras)

Collect the residuals in a vector $\hat{u} = (\hat{u}_1, \dots, \hat{u}_n)'$ and the centred fitted values in $v = (\hat{Y}_1 - \bar{Y}, \dots, \hat{Y}_n - \bar{Y})'$; then $Y - \bar{Y}\mathbf{1} = \hat{u} + v$ (part (a) in vector form). The inner product $\hat{u} \cdot v = \sum_i \hat{u}_i(\hat{Y}_i - \bar{Y}) = 0$ by the same FOC identities used above, so $\hat{u}$ and v are *orthogonal* vectors. By the Pythagorean theorem for orthogonal vectors,

$$\|Y - \bar{Y}\mathbf{1}\|^2 = \|\hat{u}\|^2 + \|v\|^2 \iff \text{TSS} = \text{SSR} + \text{ESS},$$

which is part (b) again, obtained without expanding any square. Since $\|\hat{u}\|^2 \geq 0$, the fitted (centred) vector v can never be longer than $Y - \bar{Y}\mathbf{1}$ itself, i.e. $\text{ESS} \leq \text{TSS}$ and $R^2 \leq 1$ – the geometric restatement of part (c).

Interpretation. TSS measures the total variation of Y_i around its sample mean; ESS is the share of that variation the fitted line accounts for, and SSR is the share left over in the residuals. Because residuals are a genuine "leftover" (a nonnegative sum of squares) that adds on top of ESS rather than subtracting from it, the explained share $\text{ESS}/\text{TSS} = R^2$ can never exceed 100%: a regression can explain at most all of the variation in Y , never more.

Beyond the slides. The official Tut 5 slide states the two FOC identities $\sum \hat{u}_i = 0$, $\sum \hat{u}_i \hat{Y}_i = 0$ as given (citing them as already proved in Tutorial 3 Q2) without re-deriving them; the one-line derivations above from the FOCs make this solution self-contained without requiring the reader to look them up.

Final answer

$$\boxed{\text{TSS} = \sum_{i=1}^n [\hat{u}_i + (\hat{Y}_i - \bar{Y})]^2 = \text{ESS} + \text{SSR}, \quad 0 \leq R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}} \leq 1.}$$

Question 3 (Comparing weekly working hours: SG versus HK)

Problem

Suppose that the population mean and variance of employees' working hours per week in Singapore (SG) are μ_1 and σ_1^2 , respectively, and N_1 i.i.d. sample observations were drawn from this population; the population mean and variance of employees' working hours per week in Hong Kong (HK) are μ_2 and σ_2^2 , respectively, and N_2 i.i.d. sample observations were drawn from this population.

The SG and HK random samples are totally independent. We are interested in $\mu_1 - \mu_2$, the difference between the two population means, and use $\hat{d} = \bar{Y}_1 - \bar{Y}_2$ as its estimator, where $\bar{Y}_1$ is the sample average of the N_1 SG observations and $\bar{Y}_2$ is the sample average of the N_2 HK observations.

**The formulas of means, variances, and covariances of sums of random variables can be found in Key Concept 2.3 (p.74) of "Introduction to Econometrics" or in the Supplement "Expectation, Variance, and Covariance".*

(a) Compute $E[\bar{Y}_1]$, $E[\bar{Y}_2]$, $\text{Var}[\bar{Y}_1]$, and $\text{Var}[\bar{Y}_2]$.

(b) Compute $E[\hat{d}]$, $\text{Cov}[\bar{Y}_1, \bar{Y}_2]$, and $\text{Var}[\hat{d}]$.

(Tutorial Question Set No. 5, Q3.)

Solution

Method 1: Direct expectation/variance algebra (official route)

Part (a): moments of $\bar{Y}_1$ and $\bar{Y}_2$. Let $Y_{1,1}, \dots, Y_{1,N_1}$ be the SG sample, i.i.d. with $E[Y_{1,i}] = \mu_1$, $\text{Var}(Y_{1,i}) = \sigma_1^2$, and $\bar{Y}_1 = \frac{1}{N_1} \sum_{i=1}^{N_1} Y_{1,i}$. By linearity of expectation,

$$E[\bar{Y}_1] = E\left[\frac{1}{N_1} \sum_{i=1}^{N_1} Y_{1,i}\right] = \frac{1}{N_1} \sum_{i=1}^{N_1} E[Y_{1,i}] = \frac{1}{N_1} (N_1 \mu_1) = \mu_1.$$

Because the N_1 observations are independent (i.i.d. sampling), all cross-covariance terms in $\text{Var}(\sum Y_{1,i})$ vanish, so

$$\text{Var}[\bar{Y}_1] = \text{Var}\left[\frac{1}{N_1} \sum_{i=1}^{N_1} Y_{1,i}\right] = \frac{1}{N_1^2} \sum_{i=1}^{N_1} \text{Var}(Y_{1,i}) = \frac{1}{N_1^2} (N_1 \sigma_1^2) = \frac{\sigma_1^2}{N_1}.$$

By the identical argument applied to the (independent) HK sample $Y_{2,1}, \dots, Y_{2,N_2}$,

$$\boxed{E[\bar{Y}_1] = \mu_1, \quad E[\bar{Y}_2] = \mu_2, \quad \text{Var}[\bar{Y}_1] = \frac{\sigma_1^2}{N_1}, \quad \text{Var}[\bar{Y}_2] = \frac{\sigma_2^2}{N_2}.}$$

Part (b): moments of $\hat{d} = \bar{Y}_1 - \bar{Y}_2$. *Unbiasedness.* By linearity of expectation and part (a),

$$E[\hat{d}] = E[\bar{Y}_1 - \bar{Y}_2] = E[\bar{Y}_1] - E[\bar{Y}_2] = \mu_1 - \mu_2,$$

so $\hat{d}$ is an unbiased estimator of $\mu_1 - \mu_2$.

Covariance. Every SG observation $Y_{1,i}$ is independent of every HK observation $Y_{2,j}$ (the two national samples are "totally independent" by assumption), so $\text{Cov}(Y_{1,i}, Y_{2,j}) = 0$ for all pairs (i, j) . Since covariance is bilinear,

$$\begin{aligned} \text{Cov}[\bar{Y}_1, \bar{Y}_2] &= \text{Cov}\left[\frac{1}{N_1} \sum_{i=1}^{N_1} Y_{1,i}, \frac{1}{N_2} \sum_{j=1}^{N_2} Y_{2,j}\right] \\ &= \frac{1}{N_1 N_2} \sum_{i=1}^{N_1} \sum_{j=1}^{N_2} \text{Cov}(Y_{1,i}, Y_{2,j}) = \frac{1}{N_1 N_2} \sum_{i=1}^{N_1} \sum_{j=1}^{N_2} 0 = 0. \end{aligned}$$

$$\boxed{\text{Cov}[\bar{Y}_1, \bar{Y}_2] = 0.}$$

Variance. Using $\text{Var}(A - B) = \text{Var}(A) + \text{Var}(B) - 2 \text{Cov}(A, B)$ with $A = \bar{Y}_1$, $B = \bar{Y}_2$, and the covariance result just derived,

$$\text{Var}[\hat{d}] = \text{Var}[\bar{Y}_1 - \bar{Y}_2] = \text{Var}[\bar{Y}_1] + \text{Var}[\bar{Y}_2] - 2 \text{Cov}[\bar{Y}_1, \bar{Y}_2] = \frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2} - 2(0).$$

$$\boxed{\text{Var}[\hat{d}] = \frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2}.}$$

Notice that although $\hat{d}$ is a *difference* of two estimators, the minus sign in the variance formula disappears entirely: because the two samples are independent, there is no covariance term to subtract, and the variances simply add.

Interpretation. $\hat{d} = \bar{Y}_1 - \bar{Y}_2$ is unbiased for the population gap $\mu_1 - \mu_2$ regardless of N_1, N_2 , so on average across repeated samples it recovers the true SG–HK difference in mean working hours. Its precision, however, depends on both sample sizes and both population variances: $\text{Var}(\hat{d})$ shrinks as either N_1 or N_2 grows, and independence of the two national samples is what allows the two individual variances to simply add (no covariance discount or penalty).

Beyond the slides: standard error and a large-sample confidence interval. The slides stop at $\text{Var}[\hat{d}]$; in practice σ_1^2, σ_2^2 are unknown and are replaced by the sample variances s_1^2, s_2^2 computed within each country's sample, giving the estimated standard error

$$\text{SE}(\hat{d}) = \sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}.$$

For N_1, N_2 reasonably large, $\hat{d}$ is approximately normal (by the CLT, exactly as $\bar{Y}$ was in Question 1), so a large-sample 95% confidence interval for $\mu_1 - \mu_2$ is

$$\boxed{CI_{95\%}(\mu_1 - \mu_2) = (\bar{Y}_1 - \bar{Y}_2) \pm 1.96 \text{SE}(\hat{d}) = (\bar{Y}_1 - \bar{Y}_2) \pm 1.96 \sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}.}$$

This is a preview of the difference-in-means hypothesis test and confidence interval developed formally in Chapter 6 ("Tests and Confidence Intervals" – Lecture 6's t-statistic and CI machinery applied to a two-sample comparison instead of a single regression coefficient): the same "estimator $\pm 1.96 \times$ its standard error" recipe used for $\hat{\beta}_1$ in Lecture 6 applies unchanged to $\hat{d}$, once $\text{Var}(\hat{d})$ is known.

Final answer

$$E[\bar{Y}_1] = \mu_1, \quad E[\bar{Y}_2] = \mu_2, \quad \text{Var}[\bar{Y}_1] = \frac{\sigma_1^2}{N_1}, \quad \text{Var}[\bar{Y}_2] = \frac{\sigma_2^2}{N_2}.$$

$$E[\hat{d}] = \mu_1 - \mu_2, \quad \text{Cov}[\bar{Y}_1, \bar{Y}_2] = 0, \quad \text{Var}[\hat{d}] = \frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2}.$$

Method Map

Q.	Core topic	Methods included
1	Unbiasedness, efficiency, Gauss–Markov, BLUE	(a) $\bar{Y}$ vs. $\tilde{Y} = \frac{1}{2}(Y_1 + Y_n)$ vs. Y_1 : unbiasedness + variance comparison. (b) Gauss–Markov statement, LSA1–3 + homoskedasticity, BLUE unpacked, scope of the claim. (c) $\sum a_i = 1$ condition from $E[\sum a_i Y_i] = \mu$; extension: $\sum a_i^2 = \sum (a_i - 1/n)^2 + 1/n$ minimised at $a_i = 1/n$, proving $\tilde{Y}$ is BLUE. (d) Slides 6–8 simulation explained; theoretical $\text{Var}(Y_1) = 1$ vs. $\text{Var}(\tilde{Y}) = 0.1$; pgfplots density figure.
2	$R^2 \leq 1$	Method 1: algebraic decomposition $\text{TSS} = \text{SSR} + \text{ESS}$ via $Y_i - \bar{Y} = \hat{u}_i + (\hat{Y}_i - \bar{Y})$ and the FOC identities $\sum \hat{u}_i = 0$, $\sum \hat{u}_i X_i = 0$ (re-derived). Method 2: geometric/Pythagorean orthogonality argument.
3	Two independent samples (SG vs. HK hours)	Linearity of $E[\cdot]$ and independence for $\text{Var}[\bar{Y}_1], \text{Var}[\bar{Y}_2]$; bilinearity of $\text{Cov}[\cdot, \cdot]$ to show $\text{Cov}[\bar{Y}_1, \bar{Y}_2] = 0$; $\text{Var}(A - B) = \text{Var} A + \text{Var} B - 2 \text{Cov}(A, B)$ for $\text{Var}(\hat{d})$; extension to $\text{SE}(\hat{d})$ and a large-sample 95% CI.

Technique Summary

- Unbiasedness is a statement about $E[\hat{\theta}]$; efficiency is a statement about $\text{Var}(\hat{\theta})$ given unbiasedness.** Always check both, and never rank estimators on variance alone without first confirming they are unbiased.
- ”For all μ ” forces the weight sum to 1.** Whenever $E[\sum a_i Y_i] = \mu \sum a_i$ must equal μ for every possible true mean, the only way to satisfy this identically is $\sum a_i = 1$ – a trick that reappears whenever unbiasedness is checked for a linear estimator.
- The ”spread the weights equally” identity $\sum a_i^2 = \sum (a_i - 1/n)^2 + 1/n$** is the algebraic engine behind every BLUE proof in this course (sample mean here; OLS slope/intercept in the full Gauss–Markov proof) – memorise the completing-the-square trick, not just the conclusion.
- Gauss–Markov needs LSA1–3 and homoskedasticity.** Unbiasedness of OLS survives heteroskedasticity; the BLUE (minimum-variance) claim does not, without further adjustment (GLS).
- TSS = ESS + SSR is an orthogonality fact, not a coincidence.** It follows from the two FOC identities $\sum \hat{u}_i = 0$, $\sum \hat{u}_i \hat{Y}_i = 0$, which geometrically say the residual vector is perpendicular to the (centred) fitted-value vector – hence Pythagoras, hence $R^2 \in [0, 1]$.
- Independent samples: variances add, covariances vanish.** For $\hat{d} = \bar{Y}_1 - \bar{Y}_2$ built from two independent samples, $\text{Var}(\hat{d}) = \text{Var}(\bar{Y}_1) + \text{Var}(\bar{Y}_2)$ with no cross term – the

same " ± 1.96 SE" confidence-interval recipe from single-regressor inference (Lecture 6) then applies unchanged.

Final Answers

- Q1(a)** $\bar{Y}, \tilde{Y}, Y_1$ all unbiased for μ ; $\text{Var}(\bar{Y}) = \sigma^2/n < \text{Var}(\tilde{Y}) = \sigma^2/2 < \text{Var}(Y_1) = \sigma^2$ for $n > 2$, so $\bar{Y}$ is most efficient.
- Q1(b)** Under LSA1–3 + homoskedasticity, $\hat{\beta}_0, \hat{\beta}_1$ are BLUE: smallest variance among all linear unbiased estimators of β_0, β_1 . Does not imply OLS beats every (possibly biased or nonlinear) estimator, and fails without homoskedasticity.
- Q1(c)** $\sum_{i=1}^n a_i Y_i$ unbiased for $\mu \iff \sum_{i=1}^n a_i = 1$; minimising $\text{Var} = \sigma^2 \sum a_i^2$ subject to this constraint gives $a_i = 1/n$, i.e. $\bar{Y}$ is BLUE.
- Q1(d)** $\text{Var}(Y_1) = 1$, $\text{Var}(\bar{Y}) = 0.1$; simulation (20 samples, $n = 10$) shows both estimators unbiased (averages 1.89, 1.96, both near $\mu = 2$) but $\bar{Y}$ far more concentrated (16/20 closer to 2; range $[1.16, 2.58]$ vs. $[-0.64, 4.27]$).
- Q2** $\text{TSS} = \sum [\hat{u}_i + (\hat{Y}_i - \bar{Y})]^2 = \text{ESS} + \text{SSR}$ (cross term = 0 by $\sum \hat{u}_i = 0$, $\sum \hat{u}_i X_i = 0$); hence $0 \leq R^2 = \text{ESS}/\text{TSS} = 1 - \text{SSR}/\text{TSS} \leq 1$.
- Q3** $E[\bar{Y}_1] = \mu_1$, $E[\bar{Y}_2] = \mu_2$, $\text{Var}[\bar{Y}_1] = \sigma_1^2/N_1$, $\text{Var}[\bar{Y}_2] = \sigma_2^2/N_2$; $E[\hat{d}] = \mu_1 - \mu_2$, $\text{Cov}[\bar{Y}_1, \bar{Y}_2] = 0$, $\text{Var}[\hat{d}] = \sigma_1^2/N_1 + \sigma_2^2/N_2$.